

Proof. Let $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$ be real numbers. Define

$$A = \sum_{i=1}^n b_i^2, \quad B = \sum_{i=1}^n a_i b_i, \quad C = \sum_{i=1}^n a_i^2.$$

Consider the quadratic function in $t \in \mathbb{R}$:

$$f(t) = \sum_{i=1}^n (a_i + t b_i)^2.$$

Since $f(t)$ is a sum of squares, $f(t) \geq 0$ for all t .

Expanding:

$$f(t) = \sum_{i=1}^n (a_i^2 + 2t a_i b_i + t^2 b_i^2) = \left(\sum_{i=1}^n a_i^2 \right) + 2t \left(\sum_{i=1}^n a_i b_i \right) + t^2 \left(\sum_{i=1}^n b_i^2 \right) = C + 2Bt + At^2.$$

Thus $f(t) = At^2 + 2Bt + C$.

Case 1: $A = 0$. Then all $b_i = 0$, and the inequality $(\sum a_i b_i)^2 \leq (\sum a_i^2)(\sum b_i^2)$ becomes $0 \leq 0$, which is true.

Case 2: $A > 0$. The quadratic $f(t)$ is non-negative for all t , so its discriminant must be non-positive:

$$(2B)^2 - 4AC \leq 0 \implies 4B^2 - 4AC \leq 0 \implies B^2 \leq AC.$$

Substituting back yields

$$\left(\sum_{i=1}^n a_i b_i \right)^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right),$$

which is the Cauchy-Schwarz inequality.

Equality condition: Equality holds iff the discriminant is zero, meaning $f(t)$ has a real double root t_0 . Then $f(t_0) = \sum (a_i + t_0 b_i)^2 = 0$, so $a_i + t_0 b_i = 0$ for all i , i.e., $a_i = -t_0 b_i$. Thus (a_i) and (b_i) are proportional. This also covers the $A = 0$ case. Therefore, equality holds if and only if the sequences are linearly dependent. $\square$